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GRAU DE MATEMÀTIQUES

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GRAPH NEURAL NETWORKS FOR AUTONOMOUS THEOREM PROVING

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Abstract

The main goal of this work is to prove the existence of.....

Notation: We will work over an algebraically closed field K of characteristic zero.

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Chapter 1

Generalized Algebraic Theories

1.1 Definition

We assume we have a set V of countably many distinct variables. Let W be an **alphabet**, and its elements be **symbols**.

Definition 1.1. *The set of **expressions** is defined inductively such that they are finite sequences of elements of $W \cup V \cup \{(\} \cup \{)\} \cup \{,\}$ by the clauses*

1. *if $x \in V$, then x is an expression;*
2. *if $L \in W$, then L is an expression;*
3. *if $L \in W$ and $e_1, \ldots, e_n$ are expressions, then $L(e_1, \ldots, e_n)$ is an expression.*

Definition 1.2. *A **premise** is defined to be a sequence of $V \times$ set of expressions, the empty sequence is allowed and is called the **empty premise**. A premise $((x_1, \Delta_1), \ldots, (x_n, \Delta_n))$ is written as $x_1 : \Delta_1, \ldots, x_n : \Delta_n$.*

Definition 1.3. *A **conclusion** is one of the following:*

1. *T**-conclusion** consists a single expression Δ and is written as "Δ is a type";*
2. *$\in$**-conclusion** consists of pair of expressions (t, Δ) and is written as "$t : \Delta$";*
3. *$T =$**conclusion** consists of a pair of expressions (Δ_1, Δ_2) and is written as "$\Delta_1 = \Delta_2$";*
4. *$\in =$**conclusion** consists of a triple of expressions (t_1, t_2, Δ) and is written as "$t_1 = t_2 \in \Delta$".*

Definition 1.4. *A **rule** is determined by a premise P and a conclusion C, and is written as*

$$\frac{P}{C}$$

If $\Delta, t_1, \ldots, t_n$ expressions and $x_1, \ldots, x_n$, then the expression $\Delta[t_1|x_1, \ldots, t_n|x_n]$ is the result of simultaneous replacement of $x_1, \ldots, x_n$ in Δ by $t_1, \ldots, t_n$.

Definition 1.5. *A **pretheory** consists of*

1. *a set S, of sort symbols;*
2. *a set Z, of operator symbols;*
3. *for each sort symbol A, an associated rule of the alphabet $S \cup Z$, called **introductory rule** for A, of the form*

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{A(x_1, \ldots, x_n) \text{ is a type}}$$

   for some $n \geq 0$;

4. *for each operator symbol F, an associated rule of the alphabet $S \cup Z$, called **introductory rule** for F, of the form*

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{F(x_1, \ldots, x_n) \in \Delta}$$

   for each $n \geq 0$;

5. *a set of axioms, each axiom is either a $T =$rule or a $\in$-rule of the alphabet $S \cup Z$.*

Definition 1.6. *If U is a pretheory, then*

1. *a **context** is a premise $x_1 : \Delta_1, \ldots, x_n : \Delta_n$ such that the rule*

$$\frac{x_1 : \Delta_1, \ldots, x_{n-1} : \Delta_{n-1}}{\Delta_n \text{is a type}}$$

   is a derived rule such that x_n is distinct from all of $x_1, \ldots, x_{n-1}$.

2. *TODO*

Definition 1.7. *The set of **derived rules** of U is the set of rules derivable by the following principles of derivation*

LI1 *From* $\dfrac{P}{\Delta \text{ is a type}}$ *derive* $\dfrac{P}{\Delta = \Delta}$.

LI2 *From* $\dfrac{P}{t : \Delta}$ *derive* $\dfrac{P}{t = t : \Delta}$.

LI3 *From* $\dfrac{P}{\Delta_1 = \Delta_2}$ *derive* $\dfrac{P}{\Delta_2 = \Delta_1}$.

LI4 *From* $\dfrac{P}{t_1 = t_2 : \Delta}$ *derive* $\dfrac{P}{t_2 = t_1 : \Delta}$.

LI5 *From* $\dfrac{P}{\Delta_1 = \Delta_2}$ *and* $\dfrac{P}{\Delta_2 = \Delta_3}$ *derive* $\dfrac{P}{\Delta_1 = \Delta_3}$.

LI6 *From* $\dfrac{P}{t_1 = t_2 : \Delta}$ *and* $\dfrac{P}{t_2 = t_3 : \Delta}$ *derive* $\dfrac{P}{t_1 = t_3 : \Delta}$.

LI7 *From* $\dfrac{P}{t_1 = t_2 : \Delta_1}$ *and* $\dfrac{P}{\Delta_1 = \Delta_2}$ *derive* $\dfrac{P}{t_1 = t_2 : \Delta_2}$.

T1 *From* $\dfrac{P}{\Delta_1 = \Delta_2}$ *and* $\dfrac{P}{t : \Delta_1}$ *derive* $\dfrac{P}{t : \Delta_2}$.

CF1 *For* $n \geq 0, 1 \leq i \leq n+1$*:*
From $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta_{n+1} \textit{is a type}}$ *derive* $\dfrac{x_1 : \Delta_1, \ldots, x_{n+1} : \Delta_{n+1}}{x_i : \Delta_i}$.
providing that x_{n+1} *is a variable distinct from all of* $x_1, \ldots, x_n$.

CF2(a) *For every sort symbol* A *with well-formed introductory rule*
$\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{A(x_1, \ldots, x_n) \textit{is a type}}$*, for every context* P*,*
from $\dfrac{P}{t_1 : \Delta_1}, \dfrac{P}{t_2 : \Delta[t_1|x_1]}, \ldots, \dfrac{P}{t_n : \Delta_n[t_1|x_1, \ldots, t_{n-1}|x_{n-1}]}$
derive $\dfrac{P}{A(t_1, \ldots, t_n) \textit{is a type}}$.

CF2(b) *For every operator symbol* F *with well-formed introductory rule*
$\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{F(x_1, \ldots, x_n) : \Delta}$*, for every context* P*,*
from $\dfrac{P}{t_1 : \Delta_1}, \dfrac{P}{t_2 : \Delta_2[t_1|x_1]}, \ldots, \dfrac{P}{t_n : \Delta_n[t_1|x_1, \ldots, t_{n-1}|x_{n-1}]}$
derive $\dfrac{P}{F(t_1, \ldots, t_n) : \Delta[t_1|x_1, \ldots, t_n|x_n]}$.

SI1 *If* Q *is a context, then from* $\dfrac{y_1 : \Omega_1, \ldots, y_n : \Omega_n}{\Omega = \Omega'}$ *and from*
$\dfrac{Q}{s_1 = s_2 : \Omega_1}, \dfrac{Q}{s_2 = s_2' : \Omega_2[s_1|y_1]}, \ldots, \dfrac{Q}{s_m = s_m' : \Omega_m[s_1|y_1, \ldots, s_{m-1}|y_{m-1}]}$
derive $\dfrac{Q}{\Omega[s_1|y_1, \ldots, s_m|y_m]} = \Omega'[s_1'|y_1, \ldots, s_m'|y_m]$.

SI2 *If* Q *is a context, then from* $\dfrac{y_1 : \Omega_1, \ldots, y_m : \Omega_m}{s = s' : \Omega}$ *and from*
$\dfrac{Q}{s_1 = s_1' : \Omega_1}, \dfrac{Q}{s_2 = s_2' : \Omega_2[s_1|y_1]}, \ldots, \dfrac{Q}{s_m = s_m' : \Omega_m[s_1|y_1, \ldots, s_{m-1}|y_{m-1}]}$

derive $\dfrac{Q}{s[s_1|y_1,\ldots,s_m|y_m] = s'[s'_1|y_1,\ldots,s'_m|y_m] : \Omega[s_1|y_1,\ldots,s_m|y_m]}$.

A1 *If* $\dfrac{x_1:\Delta_1,\ldots,x_n:\Delta_n}{\Delta = \Delta'}$ *is an axiom then from* $\dfrac{x_1:\Delta_1,\ldots,x_n:\Delta_n}{\Delta \textit{is a type}}$ *and* $\dfrac{x_1:\Delta_1,\ldots,x_n:\Delta_n}{\Delta'}$.

A2 *If* $\dfrac{x_1:\Delta_1,\ldots,x_n:\Delta_n}{t = t' : \Delta}$ *is an axiom, then from* $\dfrac{x_1:\Delta_1,\ldots,x_n:\Delta_n}{t:\Delta}$ *and* $\dfrac{x_1:\Delta_1,\ldots,x_n:\Delta_n}{t':\Delta}$ *derive* $\dfrac{x_1:\Delta_1,\ldots,x_n:\Delta_n}{t = t' : \Delta}$

Definition 1.8. *A pretheory is* ***well-formed*** *if all of its introductory rules and axioms are well-formed. We define a* ***Generalized Algebraic Theory*** *as a well-formed pretheory.*

1.2 The Substitution Lemma

TODO: Write this section

1.3 Models and Homomorphisms

For now on, we will consider trees to be trees with a unique least element. If Θ is a tree, and A is a node of Θ, then we will write $A \in \Theta$. Also, if B is a node of Θ and B succeeds A, then we write $A \lhd B$ *in* Θ. The least node of a tree will be denoted as 1. For any node A of a tree Θ, such that $A \neq 1$, there exists a unique $n \geq 0$, and a unique sequence $A_1,\ldots,A_n \in \Theta$ such that $1 \lhd A_1 \lhd A_2 \lhd \cdots \lhd A_n \lhd A$ in Θ.

The set of contexts of a theory is structured as a tree where the least element is the empty context and for all $n \geq 1$ the predecessor of $\langle x_1:\Delta_1,\ldots,x_n:\Delta_n\rangle$ is $\langle x_1:\Delta_1,\ldots,x_{n-1}:\Delta_{n-1}\rangle$.

The **tree of families** has as nodes the collection of sets, families of sets, families of families of sets, ... Consider $1 \lhd A_1 \lhd A_n \lhd A$ in the tree of families, then for all $a_1 \in A_1$, ..., and for all $a_n \in A_n(a_1,\ldots,a_{n-1})$, $A(a_1,\ldots,a_n)$ is a set, and we define an **operator** at A as a mapping $\lambda a_1 \in A_1 \ldots \lambda a_n \in A_n . f(a_1,\ldots,a_n)$ such that $\forall a_1 \in A_1,\ldots,\forall a_n \in A_n(a_1,\ldots,a_{n-1}), f(a_1,\ldots,a_n) \in A(a_1,\ldots,a_n)$. We will write $g(a_1,\ldots,a_n)$ for the value of the operator g with arguments $a_1,\ldots,a_n$.

When we consider a model M of some theory U, the derived rules are interpreted in M, not the expressions. The interpretation of a rule in a model M is written as that rule superscripted by M.

Consider U a theory with a model M, and R a derived T-rule of such theory $\dfrac{x_1:\Delta_1,\ldots,x_n:\Delta_n}{\Delta \text{ is a type}}$ Let R_i be the rule $\dfrac{x_1:\Delta_1,\ldots,x_{i-1}:\Delta_{i-1}}{\Delta_i \text{ is a type}}$ for $1 \leq i \leq n$.

Then R_1^M is a set, R_2^M is a R_1^M indexed family of sets, ... and we have $1 \lhd R_1^M \lhd R_2^M \lhd \cdots \lhd R_n^M \lhd R^M$. If instead we take R_t, a derived rule of U of the form $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$, we would have that R_t^M would be an operator at R^M.

Let M, M' models of U, we define a **model homomorphism** as a mapping $f : M \to M'$ such that

1. for all derived rules R of U of the form $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{is a type}}$, there is an operator f_R that for all $a_1 \in R_1^M, \ldots, a_n \in R_n^M(a_1, \ldots, a_{n-1}), a \in R^M(a_1, \ldots, a_n)$, $f_R(a_1, \ldots, a_n, a) \in R^{M'}(f_{R_1}(a_1), \ldots, f_{R_n}(a_1, \ldots, a_n))$;

2. for all derived rules R_t of the form $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$, $\forall a_1 \in R_1^M, \ldots, a_n \in R_n^M(a_1, \ldots, a_n)$,
$$f_R(a_1, \ldots, R_t^M(a_1, \ldots, a_n)) = R_t^{M'}(f_{R_1}(a_1), \ldots, f_{R_n}(a_1, \ldots, a_n))$$

3. for all derived rules of the form $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta = \Delta'}$ then $f_R = f_{R'}$ where R is $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{is a type}}$ and R' is $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta' \text{is a type}}$.

Let S be the set of sort symbols of U, to define a homomorphism we must ask for a family $\lambda A \in S.f_A$ of operators of correct types, then for every T-rule R of U, the operator f_R is defined by induction.

We define the category of models and homomorphisms of a theory U as $U - alg$.

1.4 The Category GAT

Let A_U be the alphabet of a theory U and $v_1, v_2, \ldots$ a set of variables V.

Definition 1.9. *a **preinterpretation** I of U in U' consists of a function*

$$I : A_U \mapsto Expressions\ U'$$

If I is a preinterpretation of U in U', then we define a function

$$\dot{I} : Expressions\ U \mapsto Expressions\ U'$$

by induction on the following clauses:

1. if $x \in V$, then $\dot{I}(x) = x$;

2. if $L \in A_U$, then $\dot{I}(L) = I(L)$;

3. if $L \in A_U$ and $e_1, \ldots, e_n$ are expressions in U, then

$$\dot{I}(L(e_1, \ldots, e_n)) = I(L)[\dot{I}(e_1)|v_1, \ldots, \dot{I}(e_n)|v_n]$$

Then, we can define a function

$$\hat{I} : Rules\ U \mapsto Rules\ U'$$

by

1. $\hat{I}\Big(\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}\Big) = \dfrac{x_1 : \dot{I}(\Delta_1), \ldots, x_n : \dot{I}(\Delta_n)}{\dot{I}(\Delta) \text{ is a type}}$,
2. $\hat{I}\Big(\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}\Big) = \dfrac{x_1 : \dot{I}(\Delta_1), \ldots, x_n : \dot{I}(\Delta_n)}{\dot{I}(t) : \dot{I}(\Delta)}$,
3. $\hat{I}\Big(\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta = \Delta'}\Big) = \dfrac{x_1 : \dot{I}(\Delta_1), \ldots, x_n : \dot{I}(\Delta_n)}{\dot{I}(\Delta) = \dot{I}(\Delta')}$,
4. $\hat{I}\Big(\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t = t' : \Delta}\Big) = \dfrac{x_1 : \dot{I}(\Delta_1), \ldots, x_n : \dot{I}(\Delta_n)}{\dot{I}(t) = \dot{I}(t') : \dot{I}(\Delta)}$.

Definition 1.10. *An **interpretation** I of U in U' is a preinterpretation I of U in U' such that for all introductory rules and axioms R of U, $\hat{I}(R)$ is a derived rule of U'.*

It can be shown that if I is an interpretation, then for any derived rule R of U, $\hat{I}(R)$ is a derived rule of U'.

Now we want to create an equivalence relation on the derived T-rules and $\in$-rules of a theory U which we call the **equivalence relation of identity of denotation** such that if $R \equiv R'$ by this relation, any model U should interpret R and R' as identical objects.

Semantically we should not distinguish between the rules $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$ and $\dfrac{x'_1 : \Delta, \ldots, x'_n : \Delta[x'_1|x_1, \ldots, x'_{n-1}|x_{n-1}]}{\Delta[x'_1|x_1, \ldots, x'_n|x_n] \text{ is a type}}$ as they only differ on the variable names. Similarly, we should not distinguish between $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$ and $\dfrac{x_1 : \Delta'_1, \ldots, x_n : \Delta'_n}{\Delta' \text{ is a type}}$, if for each i, $1 \leq i \leq n$, $\dfrac{x_1 : \Delta_1, \ldots, x_{i-1} : \Delta_{i-1}}{\Delta_i = \Delta'_i}$ and $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta = \Delta'}$ are both derived rules.

We define the relation $\equiv$:

- between T-rules of a theory U by

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta_{n+1} \text{ is a type}} \equiv \frac{y_1 : \Omega_1, \ldots, y_m : \Omega_m}{\Omega_{m+1} \text{ is a type}}$$

if, and only if, $n = m$ and for each $1 \leq i \leq n+1$ $\dfrac{x_1 : \Delta_1, \ldots, x_{i-1} : \Delta_{i-1}}{\Delta_i = \Omega_i[x_1|y_1, \ldots, x_{i-1}|y_{i-1}]}$ is a derived rule of U.

- between $\in$-rules of a theory U by

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta} \equiv \frac{y_1 : \Omega_1, \ldots, y_m : \Omega_m}{s : \Omega}$$

if, and only if

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta_{n+1} \text{ is a type}} \equiv \frac{y_1 : \Omega_1, \ldots, y_m : \Omega_m}{\Omega_{m+1} \text{ is a type}}$$

and $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t = s[x_1|y_1, \ldots, x_n|y_n] : \Delta}$ is a derived rule of U.

If I and J are both interpretations of a theory U in a theory U' then we say $I \equiv J$ if, and only if, for all derived T-rules and $\in$-rules R on U, $\hat{I}(R) \equiv \hat{J}(R)$. Then we define GAT as the category of Generalized Algebraic Theories and equivalence classes of interpretations.

1.5 Contexts and Realizations

Definition 1.11. *If $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ and $\langle y_1 : \Omega_1, \ldots, y_n : \Omega_m \rangle$ are contexts of a theory U, then a* ***realization*** *of $\langle y_1 : \Omega_1, \ldots, y_n : \Omega_m \rangle$ with regard to $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ is an m-tuple $\langle t_1, \ldots, t_m \rangle$ such that for each j, $1 \leq j \leq m$, the rule*

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t_j : \Omega_j[t_1|y_1, \ldots, t_{j-1}|y_{j-1}]}$$

is a derived rule of U.

Let U be a GAT, then the category $R(U)$ of contexts and realizations of U we have the morphism $\langle t_1, \ldots, t_m \rangle : \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle \to \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$ if, and only if, $\langle t_1, \ldots, t_m \rangle$ is a realization of $\langle y_1 : \Omega_1, \ldots, y_n : \Omega_m \rangle$ with regard to $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$. The identity of an object $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ is the realization $\langle x_1, \ldots, x_n \rangle$. And composition is defined as

$$\langle t_1, \ldots, t_m \rangle \circ \langle s_1, \ldots s_w \rangle = \langle s_1[t_1|y_1, \ldots, t_m|y_m], \ldots, s_w[t_1|y_1, \ldots, t_m|y_m] \rangle$$

when $\langle t_1, \ldots, t_m \rangle : \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle \to \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$ and $\langle s_1, \ldots, s_w \rangle : \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle \to \langle z_1 : \Lambda_1, \ldots, z_w : \Lambda_w \rangle$.

The set of objects of $R(U)$ is structured as a tree with the empty context $\langle \rangle$ as the least element and with any context $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ is preceded by $\langle x_1 : \Delta_1, \ldots, x_{n-1} : \Delta_{n-1} \rangle$.

For any context $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n, x_{n+1} : \Delta_{n+1}, \ldots, x_{n+m} : \Delta_{n+m} \rangle$ of U, $\langle x_1, \ldots, x_n \rangle$ is a realization of $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ with regard to $\langle x_1 : \Delta_1, \ldots, x_{n+m} : \Delta_{n+m} \rangle$. We denote this morphism as $p(\langle x_1 : \Delta_1, \ldots, x_{n+m} : \Delta_{n+m} \rangle, \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle)$. So for all $A, B \in Ob(R(U))$ such that $A \leq B$, $p(B, A)$ is defined and $p(B, A) : B \to A$ in $R(U)$.

Lemma 1.12. *Let $A = \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ and $A' = \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$, if $B = \langle y_1 : \Omega_1, \ldots, y_{m+w} : \Omega_{m+w} \rangle$ and $f = \langle t_1, \ldots, t_m \rangle$ is a realization of A' with regard to A, then*

$$\begin{tikzcd} f^*B \arrow[r, "q(f,B)"] \arrow[d, "p(f^*B,A)"'] & B \arrow[d, "p(B,A')"] \\ A \arrow[r, "f"'] & A' \end{tikzcd}$$

is a pullback in $R(U)$, where

$$f * B = \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_{m+1} : \Omega_{m+1}[t_1|y_1, \ldots, t_m|y_m], \\ \ldots, y_{m+w} : \Omega_{m+w}[t_1|y_1, \ldots, t_m|y_m] \rangle$$

$$q(f, B) = \langle t_1, \ldots, t_m, y_{m+1}, \ldots, y_{m+w} \rangle$$

This motivates the following definition.

Chapter 2

títol del capítol 2